MAT2002 HW3 文杰 123090612

1

The characteristic equation is

$$r^2 + 2r + 2 = 0.$$

Its roots are

$$r = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i.$$

The general solution is then

$$y(t) = e^{-t}\big(c_1 \cos t + c_2 \sin t\big).$$

Differentiating the general solution gives:

$$\begin{aligned} y'(t) &= -e^{-t}(c_1 \cos t + c_2 \sin t) + e^{-t}(-c_1 \sin t + c_2 \cos t) \\ &= e^{-t}\Big[(-c_1 + c_2)\cos t + (-c_1 - c_2)\sin t\Big]. \end{aligned}$$

Using $y(\pi/4) = 2$:

$$e^{-\pi/4}\left(c_1 \frac{\sqrt{2}}{2} + c_2 \frac{\sqrt{2}}{2}\right) = 2$$

$$\therefore c_1 + c_2 = 2\sqrt{2}e^{\pi/4}.$$

Using $y'(\pi/4) = -2$:

$$e^{-\pi/4}\left[(-c_1 + c_2)\frac{\sqrt{2}}{2} + (-c_1 - c_2)\frac{\sqrt{2}}{2}\right] = -2.$$

$$\therefore c_2 = \sqrt{2}e^{\pi/4}.$$

Therefore, the specific solution is

$$\boxed{y(t) = \sqrt{2}e^{\pi/4 - t}(\cos t + \sin t).}$$

2

The characteristic equation is

$$r^2 + 1 = 0,$$

so the roots are

$$r = \pm i.$$

Thus $\lambda = 0,\ \mu = 1$, and the general solution is

$$y(t) = c_1 \cos t + c_2 \sin t.$$

Differentiating gives:

$$y'(t) = -c_1 \sin t + c_2 \cos t.$$

Applying the initial conditions at $t = \frac{\pi}{3}$:

$$y\left(\frac{\pi}{3}\right) = \frac{1}{2}c_1 + \frac{\sqrt{3}}{2}c_2 = 2. = 2$$

Also,

$$y'\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}c_1 + \frac{1}{2}c_2 = -4$$

$$\therefore c_1 = 1 + 2\sqrt{3}, \quad c_2 = \sqrt{3} - 2.$$

Therefore, the solution to the IVP is

$$\boxed{y(t) = (1 + 2\sqrt{3})\cos t + (\sqrt{3} - 2)\sin t.}$$

3

(i) $t^2y'' + 3ty' - 3y = 0$

Assume

$$y = t^r.$$

Then

$$y' = rt^{r-1}, \qquad y'' = r(r-1)t^{r-2}.$$

Substituting into the ODE:

$$t^2[r(r-1)t^{r-2}] + 3t[rt^{r-1}] - 3t^r = 0.$$

So

$$\big(r(r-1) + 3r - 3\big)t^r = 0.$$

Hence the indicial equation is

$$r^2 + 2r - 3 = (r-1)(r+3) = 0$$

Thus the roots are

$$r_1 = 1, \qquad r_2 = -3.$$

Therefore the general solution is

$$\boxed{y(t) = c_1 t + c_2 t^{-3}}, \qquad t > 0.$$

(ii) $t^2y'' + 7ty' + 10y = 0$

Let

$$y = t^r.$$

Then

$$y' = rt^{r-1}, \qquad y'' = r(r-1)t^{r-2}.$$

Substituting:

$$t^2[r(r-1)t^{r-2}] + 7t[rt^{r-1}] + 10t^r = 0.$$

So

$$\big(r(r-1) + 7r + 10\big)t^r = 0.$$

Hence

$$r(r-1) + 7r + 10 = r^2 + 6r + 10 = 0$$

Solving for r:

$$r = \frac{-6 \pm \sqrt{36 - 40}}{2} = \frac{-6 \pm 2i}{2} = -3 \pm i.$$

For complex roots $r = \lambda \pm i\mu$, the Euler-equation solution is

$$y(t) = t^{\lambda}\big(c_1 \cos(\mu \ln t) + c_2 \sin(\mu \ln t)\big), \quad t > 0,$$

which follows from $t^{\lambda + i\mu} = t^{\lambda} e^{i\mu \ln t}$.

Here $\lambda = -3,\ \mu = 1$. Therefore

$$\boxed{y(t) = t^{-3}\big(c_1 \cos(\ln t) + c_2 \sin(\ln t)\big)}, \qquad t > 0.$$

4

(i) $9y'' + 6y' + 82y = 0, \quad y(0) = -1, \quad y'(0) = 2$

Compute the roots:

$$r = \frac{-6 \pm \sqrt{-2916}}{18} = \frac{-6 \pm 54i}{18} = -\frac{1}{3} \pm 3i.$$

Hence the general solution is

$$y(t) = e^{-t/3}\big(c_1 \cos 3t + c_2 \sin 3t\big).$$

From $y(0) = -1$:

$$c_1 = -1.$$

Differentiate:

$$y'(t) = e^{-t/3}\left[-\frac{1}{3}(c_1\cos 3t + c_2\sin 3t) - 3c_1\sin 3t + 3c_2\cos 3t\right].$$

So

$$y'(0) = -\frac{1}{3}c_1 + 3c_2 = 2.$$

Substitute $c_1 = -1$:

$$c_2 = \frac{5}{9}.$$

Therefore,

$$\boxed{y(t) = e^{-t/3}\left(-\cos 3t + \frac{5}{9}\sin 3t\right).}$$

(ii) $y'' + 4y' + 4y = 0, \quad y(-1) = 2, \quad y'(-1) = 1$

The characteristic equation is

$$r^2 + 4r + 4 = 0 = (r+2)^2,$$

so there is a repeated root $r = -2$.

The general solution is

$$y(t) = c_1e^{-2t} + c_2te^{-2t}.$$

Differentiate:

$$y'(t) = -2c_1e^{-2t} + c_2e^{-2t} - 2c_2te^{-2t} = e^{-2t}\big(-2c_1 + c_2 - 2c_2t\big).$$

Apply $t = -1$:

From $y(-1) = 2$:

$$c_1e^2 - c_2e^2 = 2 \quad \Rightarrow \quad c_1 - c_2 = 2e^{-2}.$$

From $y'(-1) = 1$:

$$e^2(-2c_1 + 3c_2) = 1 \quad \Rightarrow \quad -2c_1 + 3c_2 = e^{-2}.$$

$$\therefore c_1 = 7e^{-2}, \quad c_2 = 5e^{-2}.$$

Thus

$$\boxed{y(t) = 7e^{-2}e^{-2t} + 5e^{-2}te^{-2t} = e^{-2t-2}(7+5t).}$$

5

Each ODE is written in standard form

$$y'' + P(x)y' + Q(x)y = 0,$$

then the reduction-of-order formula is used

$$\boxed{y_2 = y_1\int \frac{e^{-\int P(x)dx}}{y_1^2}dx}$$

(i) $xy'' - y' + 4x^3y = 0, \quad x > 0; \quad y_1(x) = \sin x^2$

First divide by x:

$$y'' - \frac{1}{x}y' + 4x^2y = 0.$$

So

$$P(x) = -\frac{1}{x}.$$

Using the formula,

$$y_2 = y_1\int \frac{e^{-\int P(x)dx}}{y_1^2}dx.$$

Compute:

$$-\int P(x)dx = -\int\left(-\frac{1}{x}\right)dx = \int\frac{1}{x}dx = \ln x$$

since $x > 0$. Hence

$$e^{-\int P(x)dx} = e^{\ln x} = x.$$

Because $y_1 = \sin x^2$,

$$y_2 = \sin x^2 \int \frac{x}{\sin^2 x^2} dx = \sin x^2 \int x \csc^2(x^2) dx.$$

Let

$$u = x^2, \qquad du = 2x dx.$$

Then

$$\int x \csc^2(x^2) dx = \frac{1}{2} \int \csc^2 u du = -\frac{1}{2} \cot u + C = -\frac{1}{2} \cot(x^2) + C.$$

Therefore

$$y_2 = \sin x^2 \left(-\frac{1}{2} \cot x^2 \right) = -\frac{1}{2} \cos x^2.$$

So a second linearly independent solution is

$$\boxed{y_2(x) = \cos x^2}$$

(up to a nonzero constant multiple).

(ii) $(x-1)y'' - xy' + y = 0, \quad x > 1; \quad y_1(x) = e^x$

First divide by $x - 1$:

$$y'' - \frac{x}{x-1} y' + \frac{1}{x-1} y = 0.$$

Thus

$$P(x) = -\frac{x}{x-1}.$$

Again,

$$y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx.$$

Compute

$$-\int P(x) dx = -\int \left(-\frac{x}{x-1} \right) dx = \int \frac{x}{x-1} dx.$$

Now

$$\frac{x}{x-1} = 1 + \frac{1}{x-1},$$

so

$$\int \frac{x}{x-1} dx = x + \ln(x-1)$$

since $x > 1$. Therefore

$$e^{-\int P(x) dx} = e^{x + \ln(x-1)} = e^x (x-1).$$

Since $y_1 = e^x$,

$$y_2 = e^x \int \frac{e^x (x-1)}{e^{2x}} dx = e^x \int (x-1) e^{-x} dx.$$

Now integrate:

$$\int (x-1) e^{-x} dx = -x e^{-x} + C$$

because

$$\frac{d}{dx} (-x e^{-x}) = (-1 + x) e^{-x} = (x-1) e^{-x}.$$

Hence

$$y_2 = e^x (-x e^{-x}) = -x.$$

So a second linearly independent solution is

$$\boxed{y_2(x) = x}$$

(up to a nonzero constant multiple; $-x$ is equivalent).

6

Rewrite the ODE:

$$y'' - y' - 2y = \frac{1}{2}e^{2t} + \frac{1}{2}e^{-2t}.$$

Complementary solution

Solve the homogeneous equation

$$y'' - y' - 2y = 0.$$

Its characteristic equation is

$$r^2 - r - 2 = 0,$$

Hence the roots are

$$r = 2, -1.$$

Therefore,

$$y_c(t) = c_1 e^{2t} + c_2 e^{-t}.$$

Particular solution

For $\frac{1}{2}e^{2t}$

Try

$$Y_1(t) = Ate^{2t}.$$

Compute derivatives:

$$Y_1' = A(1 + 2t)e^{2t},$$
$$Y_1'' = A(4 + 4t)e^{2t}.$$

Substitute into the left-hand side:

$$Y_1'' - Y_1' - 2Y_1 = A(4 + 4t)e^{2t} - A(1 + 2t)e^{2t} - 2Ate^{2t} = 3Ae^{2t}.$$

It is required that

$$3Ae^{2t} = \frac{1}{2}e^{2t},$$

so

$$A = \frac{1}{6}.$$

Thus

$$Y_1(t) = \frac{1}{6}te^{2t}.$$

For $\frac{1}{2}e^{-2t}$

Since -2 is **not** a root of the characteristic equation, try

$$Y_2(t) = Be^{-2t}.$$

Then

$$Y_2' = -2Be^{-2t}, \qquad Y_2'' = 4Be^{-2t}.$$

Substitute:

$$Y_2'' - Y_2' - 2Y_2 = 4Be^{-2t} - (-2Be^{-2t}) - 2Be^{-2t} = 4Be^{-2t}.$$

It is required that

$$4Be^{-2t} = \frac{1}{2}e^{-2t},$$

so

$$B = \frac{1}{8}.$$

Thus

$$Y_2(t) = \frac{1}{8}e^{-2t}.$$

So a particular solution is

$$Y(t) = \frac{1}{6}te^{2t} + \frac{1}{8}e^{-2t}.$$

General solution

Hence the general solution is

$$\boxed{y(t) = c_1 e^{2t} + c_2 e^{-t} + \frac{1}{6} t e^{2t} + \frac{1}{8} e^{-2t}}.$$

7

For the homogeneous part,

$$y'' - 2y' - 3y = 0.$$

Its characteristic equation is

$$r^2 - 2r - 3 = 0 = (r-3)(r+1),$$

so the roots are $r = 3, -1$. Hence the complementary solution is

$$y_c(t) = c_1 e^{3t} + c_2 e^{-t}.$$

Now find a particular solution for the forcing term $3te^{2t}$. Since the right-hand side is $P_1(t)e^{2t}$ and 2 is not a root of the characteristic equation, the standard undetermined-coefficients guess for a degree-1 polynomial times an exponential is:

$$Y(t) = e^{2t}(At + B).$$

Compute the derivatives:

$$Y'(t) = e^{2t}(2At + 2B + A),$$
$$Y''(t) = e^{2t}(4At + 4B + 4A).$$

Substitute into the ODE:

$$Y'' - 2Y' - 3Y = e^{2t}\Big[(4At + 4B + 4A) - 2(2At + 2B + A) - 3(At + B)\Big] = e^{2t}(-3At + 2A - 3B)$$

Match coefficients with $3te^{2t}$:

$$-3A = 3, \qquad 2A - 3B = 0.$$
$$\therefore A = -1, \qquad B = -\frac{2}{3}.$$
$$\therefore Y(t) = e^{2t}\left(-t - \frac{2}{3}\right).$$

Therefore the general solution is

$$y(t) = c_1 e^{3t} + c_2 e^{-t} - \left(t + \frac{2}{3}\right)e^{2t}.$$

Now apply the initial conditions.

From $y(0) = 1$,

$$c_1 + c_2 - \frac{2}{3} = 1 \quad \Rightarrow \quad c_1 + c_2 = \frac{5}{3}.$$

Differentiate:

$$y'(t) = 3c_1 e^{3t} - c_2 e^{-t} - e^{2t}\left(2t + \frac{7}{3}\right).$$

From $y'(0) = 0$,

$$3c_1 - c_2 - \frac{7}{3} = 0 \quad \Rightarrow \quad 3c_1 - c_2 = \frac{7}{3}.$$
$$\therefore c_1 = 1, \quad c_2 = \frac{2}{3}.$$

Hence the solution to the IVP is

$$\boxed{y(t) = e^{3t} + \frac{2}{3}e^{-t} - \left(t + \frac{2}{3}\right)e^{2t}}.$$

Verification:

$$y(0) = 1 + \frac{2}{3} - \frac{2}{3} = 1, \qquad y'(0) = 3 - \frac{2}{3} - \frac{7}{3} = 0.$$

The final answer is

$$\boxed{y(t) = e^{3t} + \frac{2}{3}e^{-t} - \left(t + \frac{2}{3}\right)e^{2t}.}$$

8

(i) Show that v satisfies $v'' - 5v' = 2$

Let

$$Y(t) = v(t)e^{-t}.$$

Then

$$Y'(t) = v'(t)e^{-t} - v(t)e^{-t} = e^{-t}(v' - v),$$

and

$$Y''(t) = \frac{d}{dt}\big(e^{-t}(v' - v)\big) = e^{-t}(v'' - 2v' + v).$$

Substitute Y, Y', Y'':

$$e^{-t}(v'' - 2v' + v) - 3e^{-t}(v' - v) - 4e^{-t}v = 2e^{-t}.$$

Factor out e^{-t}:

$$e^{-t}\Big[(v'' - 2v' + v) - 3(v' - v) - 4v\Big] = e^{-t}(v'' - 5v') = 2e^{-t}.$$

Since $e^{-t} \neq 0$,

$$\boxed{v'' - 5v' = 2.}$$

(ii) Let $w = v'$. Show that w satisfies $w' - 5w = 2$, and solve for w

Set

$$w(t) = v'(t).$$

Then $w'(t) = v''(t)$, so the equation becomes

$$\boxed{w' - 5w = 2.}$$

This is a first-order linear ODE. Using the integrating-factor method for equations of the form $y' + p(t)y = q(t)$, an integrating factor is $e^{\int p(t)dt}$.

Here

$$w' - 5w = 2 \quad \Longrightarrow \quad p(t) = -5.$$

So an integrating factor is

$$\mu(t) = e^{\int -5dt} = e^{-5t}.$$

Multiply both sides by e^{-5t}:

$$e^{-5t}w' - 5e^{-5t}w = 2e^{-5t}.$$

Thus

$$\frac{d}{dt}\big(e^{-5t}w\big) = 2e^{-5t}.$$

Integrate:

$$e^{-5t}w = \int 2e^{-5t}dt = -\frac{2}{5}e^{-5t} + c_1.$$

Multiply by e^{5t}:

$$\boxed{w(t) = v'(t) = -\frac{2}{5} + c_1e^{5t}.}$$

(iii) Integrate w to find v, then find $Y(t)$

Since

$$v'(t) = -\frac{2}{5} + c_1e^{5t},$$

integrate:

$$v(t) = \int \left(-\frac{2}{5} + c_1e^{5t}\right)dt = -\frac{2}{5}t + \frac{1}{5}c_1e^{5t} + c_2.$$

Therefore

$$Y(t) = v(t)e^{-t} = \left(-\frac{2}{5}t + \frac{1}{5}c_1e^{5t} + c_2\right)e^{-t}.$$

Distribute e^{-t}:

$$Y(t) = -\frac{2}{5}te^{-t} + \frac{1}{5}c_1e^{4t} + c_2e^{-t}.$$

So

$$\boxed{Y(t) = -\frac{2}{5}te^{-t} + \frac{1}{5}c_1e^{4t} + c_2e^{-t}.}$$

Hence the desired particular solution is the first term:

$$\boxed{Y_p(t) = -\frac{2}{5}te^{-t}.}$$

9

First, the homogeneous equation is

$$y'' + \lambda^2 y = 0.$$

Its characteristic equation is

$$r^2 + \lambda^2 = 0,$$

so the roots are $r = \pm i\lambda$. Hence

$$y_c(t) = c_1\cos(\lambda t) + c_2\sin(\lambda t).$$

Now, a particular solution is found for each term $a_m \sin(m\pi t)$.
For fixed m, let

$$Y_m(t) = A_m \sin(m\pi t).$$

Then

$$Y_m''(t) = -(m\pi)^2 A_m \sin(m\pi t),$$

so

$$Y_m'' + \lambda^2 Y_m = \big(\lambda^2 - (m\pi)^2\big)A_m \sin(m\pi t).$$

To make this equal to $a_m \sin(m\pi t)$, it is required that

$$\big(\lambda^2 - (m\pi)^2\big)A_m = a_m.$$

Since $\lambda \neq m\pi$, the denominator is nonzero, so

$$A_m = \frac{a_m}{\lambda^2 - (m\pi)^2}.$$

Thus

$$Y_m(t) = \frac{a_m}{\lambda^2 - (m\pi)^2}\sin(m\pi t).$$

Summing over $m = 1, \ldots, N$, a particular solution is

$$Y(t) = \sum_{m=1}^{N} \frac{a_m}{\lambda^2 - (m\pi)^2}\sin(m\pi t).$$

Therefore, the general solution is

$$\boxed{y(t) = c_1\cos(\lambda t) + c_2\sin(\lambda t) + \sum_{m=1}^{N} \frac{a_m}{\lambda^2 - (m\pi)^2}\sin(m\pi t)}$$

with arbitrary constants $c_1, c_2 \in \mathbb{R}$.

10

(i) $t^2y'' - 2y = 3t^2 - 1, \quad t > 0, \quad y_1 = t^2,\ y_2 = t^{-1}$

The corresponding homogeneous equation is $t^2y'' - 2y = 0$.

Substitute $y_1(t) = t^2$:
$y_1'(t) = 2t$, $y_1''(t) = 2$,
$t^2 \cdot 2 - 2 \cdot t^2 = 2t^2 - 2t^2 = 0$.

Substitute $y_2(t) = t^{-1}$:
$y_2'(t) = -t^{-2}$, $y_2''(t) = 2t^{-3}$,
$t^2 \cdot 2t^{-3} - 2 \cdot t^{-1} = 2t^{-1} - 2t^{-1} = 0$.

Both satisfy the homogeneous equation.

The equation in standard form is

$$y'' - \frac{2}{t^2}y = 3 - t^{-2}.$$

The Wronskian is

$$W(y_1, y_2) = t^2(-t^{-2}) - t^{-1}(2t) = -3.$$

Then

$$u_1' = -\frac{y_2 g}{W} = \frac{1}{t} - \frac{1}{3}t^{-3}, \quad u_1 = \ln t + \frac{1}{6t^2},$$
$$u_2' = \frac{y_1 g}{W} = -t^2 + \frac{1}{3}, \quad u_2 = -\frac{t^3}{3} + \frac{t}{3}$$

(omitting constants of integration). A particular solution is

$$\boxed{y_p = u_1 y_1 + u_2 y_2 = t^2 \ln t - \frac{1}{3}t^2 + \frac{1}{2}.}$$

(ii) $ty'' - (1+t)y' + y = t^2 e^{2t}, \quad t > 0, \quad y_1 = 1 + t,\ y_2 = e^t$

The corresponding homogeneous equation is $ty'' - (1+t)y' + y = 0$.

Substitute $y_1(t) = 1 + t$:
$y_1'(t) = 1$, $y_1''(t) = 0$,
$t \cdot 0 - (1+t) \cdot 1 + (1+t) = -1 - t + 1 + t = 0$.

Substitute $y_2(t) = e^t$:
$y_2'(t) = e^t$, $y_2''(t) = e^t$,
$te^t - (1+t)e^t + e^t = e^t[t - 1 - t + 1] = 0$.

Both satisfy the homogeneous equation.

The equation in standard form is

$$y'' - \frac{1+t}{t}y' + \frac{1}{t}y = te^{2t}.$$

The Wronskian is

$$W(y_1, y_2) = (1+t)e^t - e^t \cdot 1 = te^t.$$

Then

$$u_1' = -\frac{y_2 g}{W} = -e^{2t}, \quad u_1 = -\frac{1}{2}e^{2t},$$
$$u_2' = \frac{y_1 g}{W} = (1+t)e^t, \quad u_2 = te^t$$

(omitting constants of integration). A particular solution is

$$\boxed{y_p = u_1 y_1 + u_2 y_2 = \frac{1}{2}(t-1)e^{2t}.}$$